YAKEEN

NEET 2.0 2027

Physics | Lecture - 5

Vector

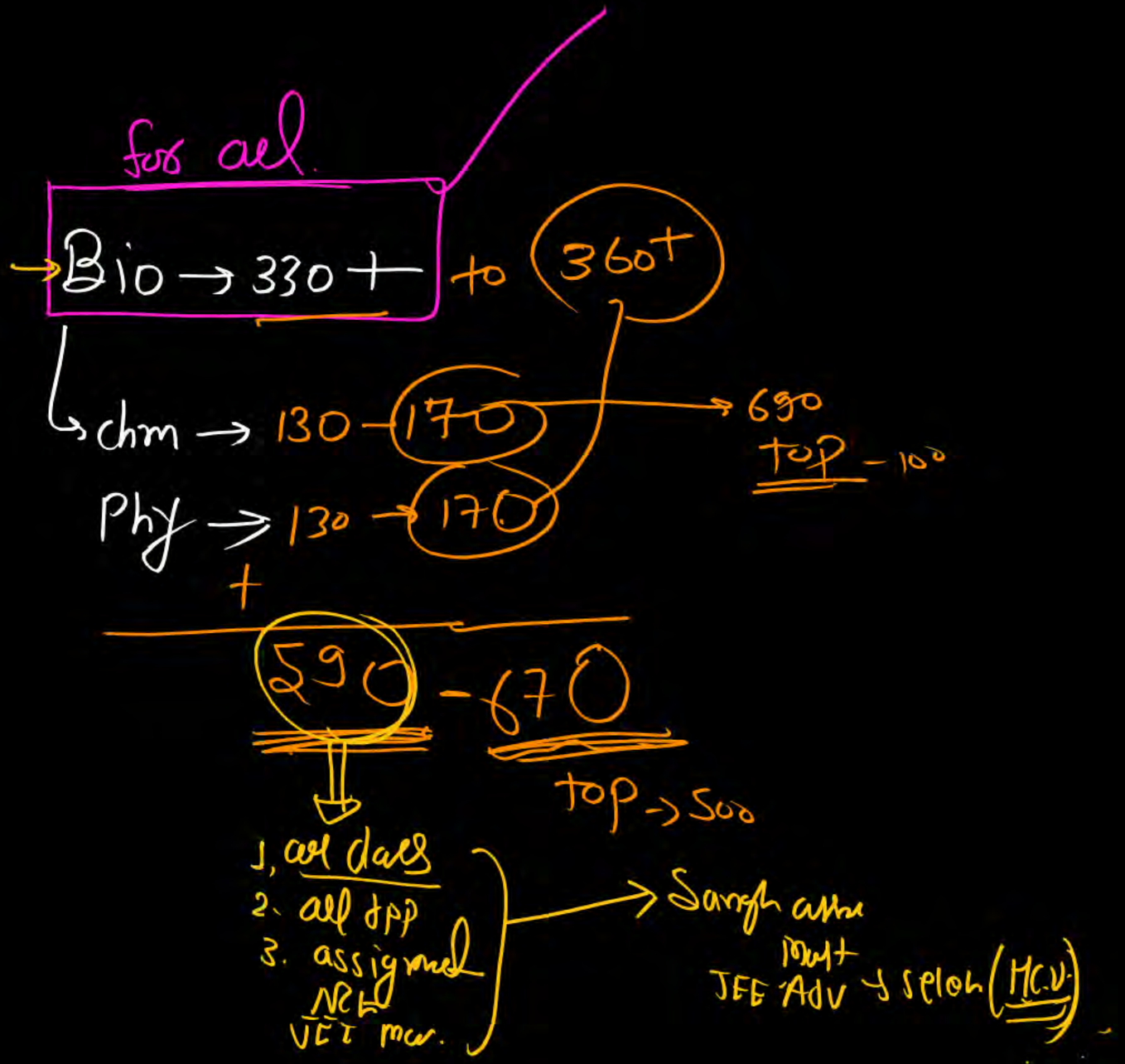
By- Manish Raj (MR Sir)



Todays Goal

⇒ Vector in Physics *

⇒ dot and cross Product ✓


for all
Bio → 330+
to
360+
chm → 130 – 170
Phy → 130 – 170
690
top – 100
+
590 – 670
top → 500
1. all data
2. all opp
3. assigned
NCL
VIT mar.
Sarafh algo
result
JEE Adv 4 selon (M.C.V.)

Question

Find value of β if $\vec{A}$ is *a* unit vector $\vec{A} = 0.4\hat{i} + 0.3\hat{j} + \beta\hat{k}$.

magnitude of unit vector is 1

$$\begin{cases} |\vec{A}| = 1 \\ \vec{A} \cdot \vec{A} = 1 \end{cases}$$

$$|\vec{A}| = \sqrt{(0.4)^2 + (0.3)^2 + \beta^2}$$

$$1^2 = \frac{16}{100} + \frac{9}{100} + \beta^2$$

$$1 - 0.16 - 0.09 = \beta^2$$

$$\beta^2 = 1 - 0.25$$

$$\beta^2 = 0.75$$

$$\beta = \sqrt{0.75}$$

Question

If $\vec{A} = 2\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{B} = \hat{i} + \hat{j} + 2\hat{k}$ then find vector $\vec{C}$ magnitude same as $\vec{A}$ and direction along $\vec{B}$.

$$\vec{C} = C\,\hat{C}$$

$$\vec{C} = A\,\hat{B} = A\,\frac{\vec{B}}{|\vec{B}|}$$

$$= 3\,\frac{\vec{B}}{B}$$

$$|\vec{A}| = \sqrt{2^2+1^2+2^2}$$

$$= \sqrt{9} = 3$$

$$|\vec{B}| = \sqrt{1^2+1^2+2^2}$$

$$= \sqrt{6}$$

$$\vec{C} = \frac{3\,(\hat{i}+\hat{j}+2\hat{k})}{\sqrt{6}}$$

Question Copy

Force of magnitude 20 N acting along vector $\vec{A} = 2\hat{i} + 3\hat{j} - \sqrt{3}\hat{k}$, then find force in vector force and acceleration of object if mass 5 kg.

Parallel

$|\vec{F}| = 20N$

Parallel $\rightarrow$ $\hat{F} = \hat{A}$

$\vec{F} = |\vec{F}|\,\hat{F}$

$= 20\,[\hat{A}] = 20\left(\frac{\vec{A}}{A}\right) = \frac{20\,(2i+3\hat{j}-\sqrt{3}\hat{k})}{\sqrt{2^2+3^2+(-\sqrt{3})^2}}$

$= \frac{20\,(2i+3j-\sqrt{3}\hat{k})}{\sqrt{16}}$

$= 5(2i+3\hat{j}-\sqrt{3}\hat{k})$

$a = \frac{\vec{F}}{m} = 2i+3\hat{j}-\sqrt{3}\hat{k}$

Question

No copy!

Three forces given by vectors $2\hat{i} + 2\hat{j}$; $-2\hat{i} - 2\hat{j}$ and $-4\hat{i}$ are acting together on a point object at rest. The object moves along the direction

1. x-axis
2. y-axis
3. z-axis
4. Object does not move

$\vec{F}_{net} = 2\hat{i} + 2\hat{j} - 2\hat{i} - 2\hat{j} - 4\hat{i}$

$\vec{F}_{net} = -4\hat{i}$

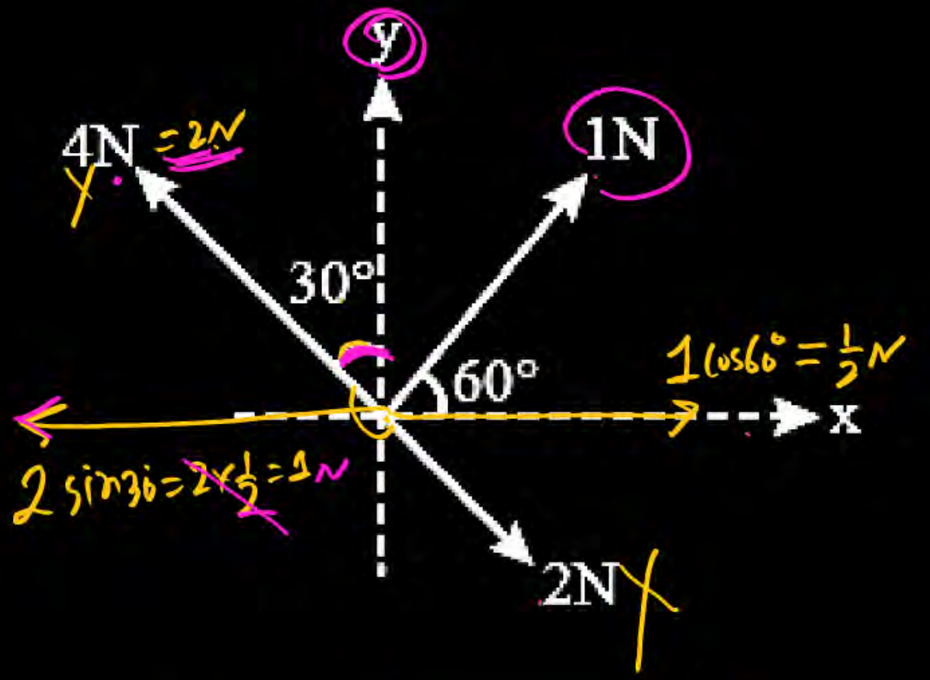
Question NEET/JEE CPY

Three force acting on the body as shown in figure. To have resultant force only along y-axis, magnitude of minimum additional force needed is?

1. $\sqrt{3}$ N
2. $\frac{\sqrt{3}}{2}$ N
3. 1.5 N
4. $\frac{1}{2}$ N

$F_{net} = Y(\hat{j})$

$(F_{net})_{x=0} = 0N$





Question

Two vector $\vec{A} = 6\hat{i}$ and $\vec{B} = 8\hat{j}$ then find $\vec{A} + \vec{B}$, direction of $\vec{A} + \vec{B}$ from $\vec{A}$.

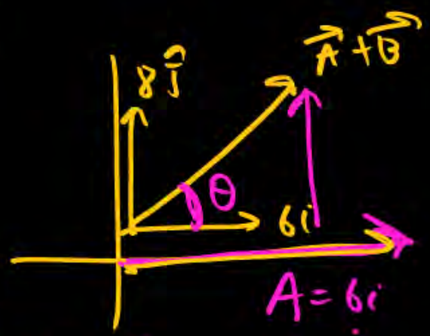
Ans

$\vec{A} = 6\hat{i}$

$\vec{B} = 8\hat{j}$

$\vec{A} + \vec{B} = 6\hat{i} + 8\hat{j} = \vec{C}$ (let)

$\vec{A} = 6\hat{i}$



$\tan\theta = \frac{8}{6} = \frac{4}{3}$

$\theta = 53°$

Method

$\vec{A} \cdot \vec{C} = AC\cos\theta$

$36 + 0 = 6\sqrt{6^2+8^2}\cos\theta$

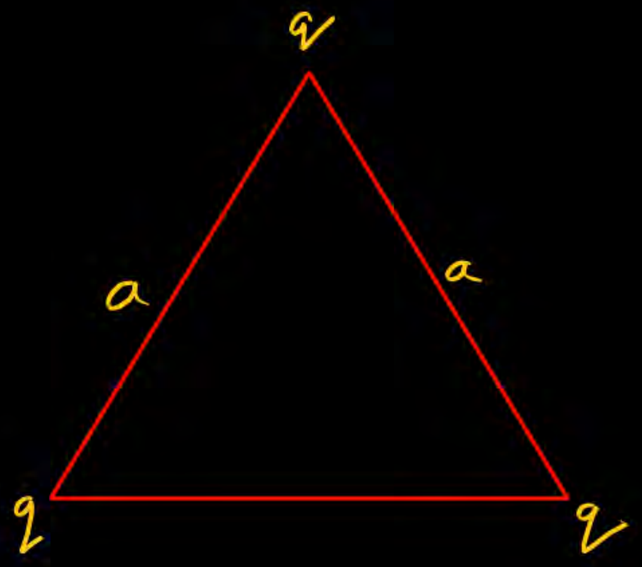
$36 = 6 \times \sqrt{100}\cos\theta$

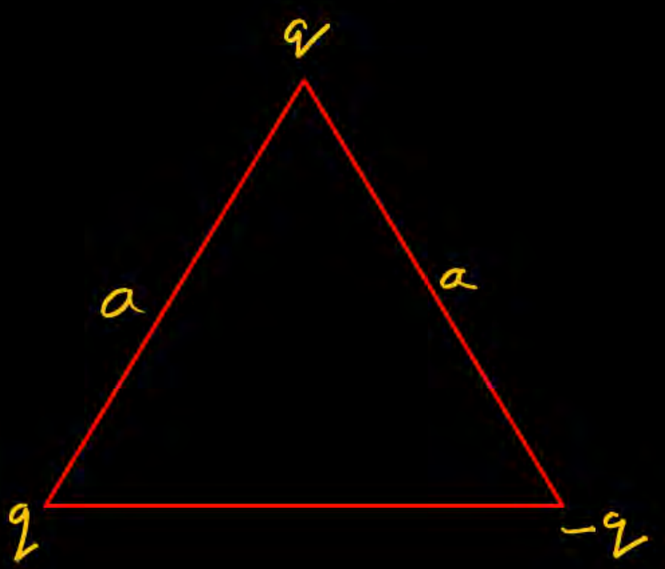
$\cos\theta = \frac{36}{6\times 10} = \frac{6}{10} = \frac{3}{5}$

$\cos\theta = \frac{3}{5}$ ✓

$\theta = 53°$ ✓

find force on any one charge due to other two



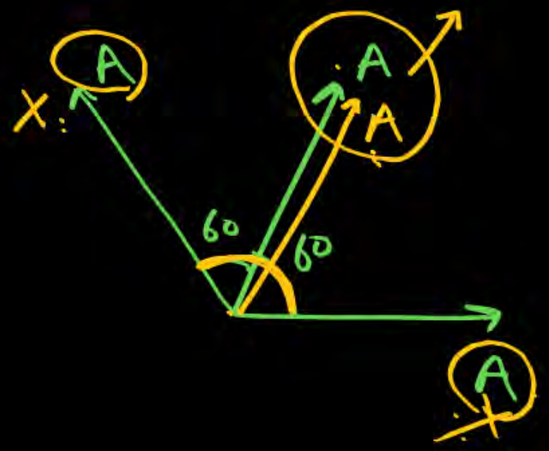


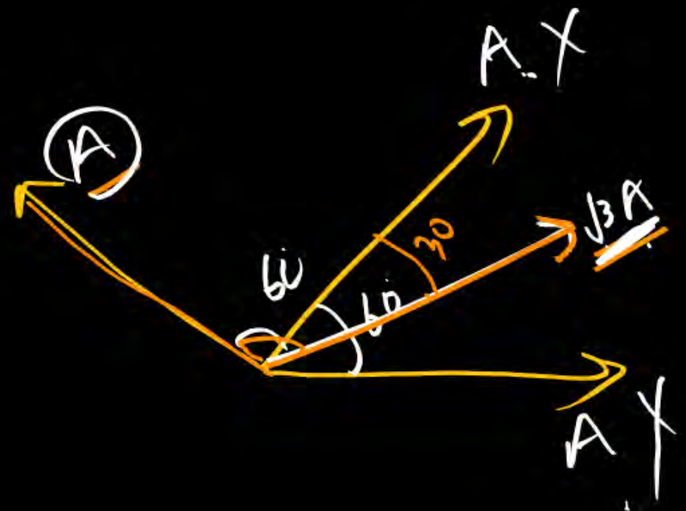
Question

copy

Three vectors each of magnitude A are acting at a point such that angle between any two consecutive vectors in same plane is 60°. The magnitude of their resultant is

1. $2A$
2. $\sqrt{2}A$
3. $\sqrt{3}A$
4. $\sqrt{6}A$

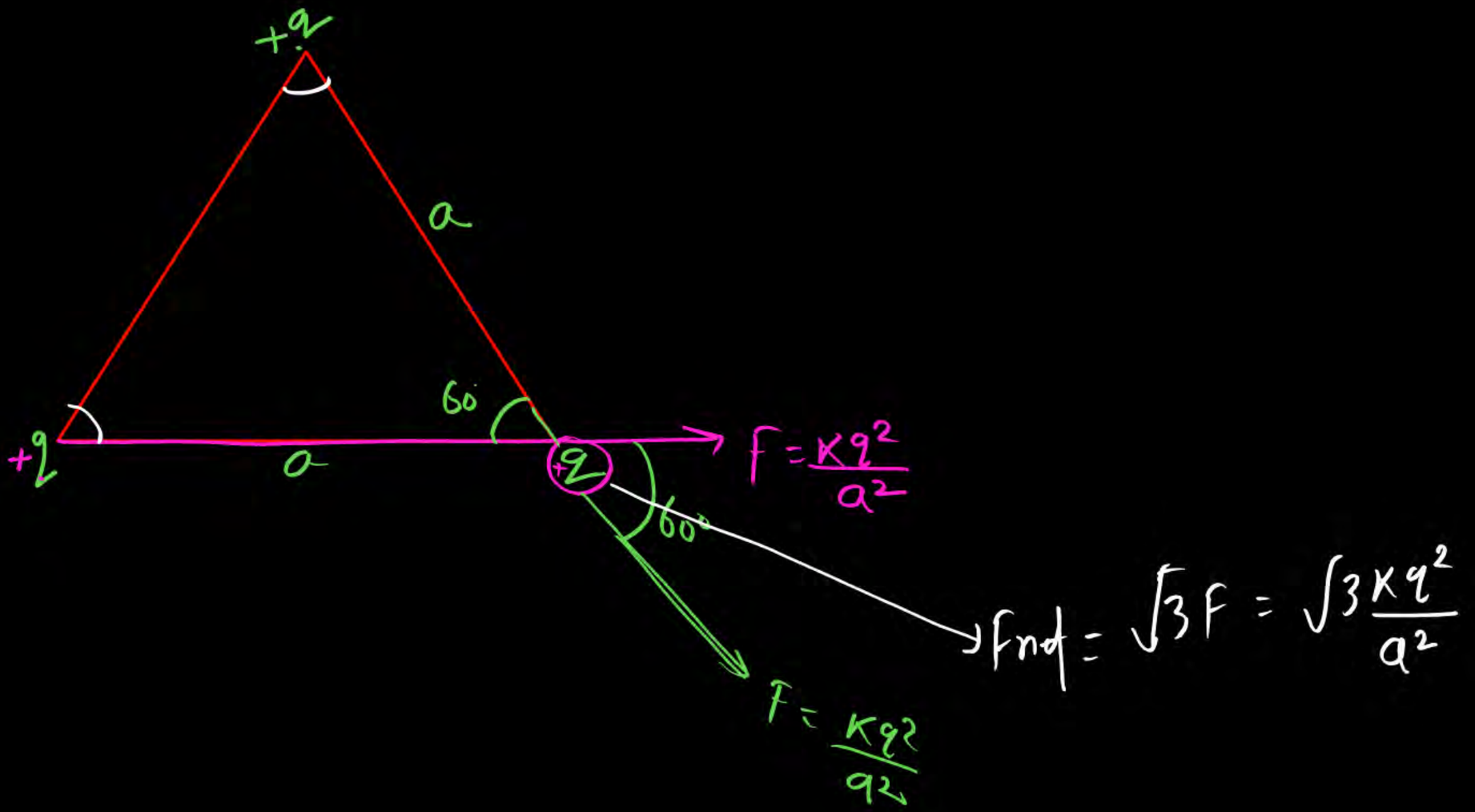




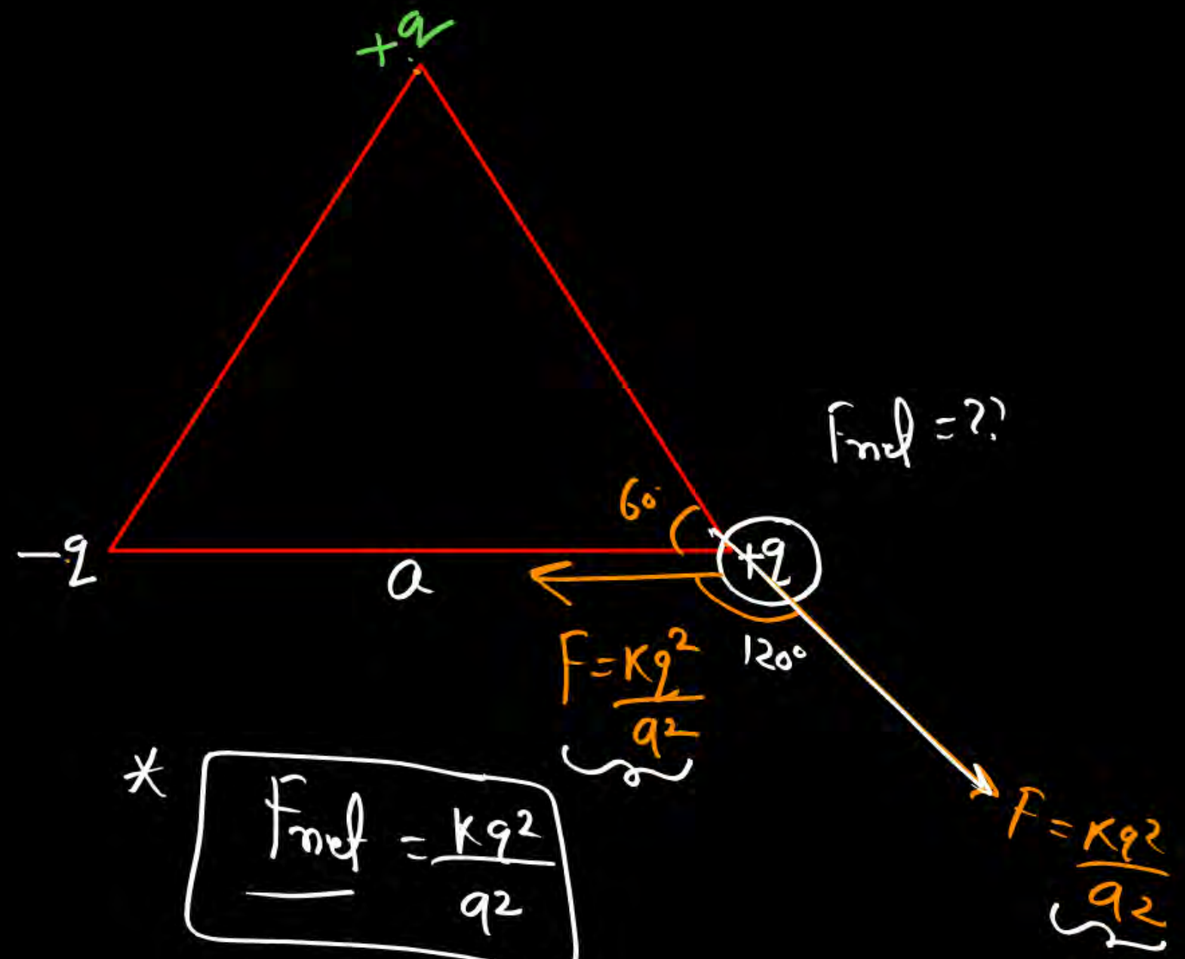
$$|\vec{R}| = \sqrt{(\sqrt{3}A)^2 + A^2 + 2A(\sqrt{3}A)\cos 30^\circ}$$

$$= \sqrt{3A^2 + A^2} = \sqrt{4A^2} = 2A$$

find force on any-one charge due to other 2-charge



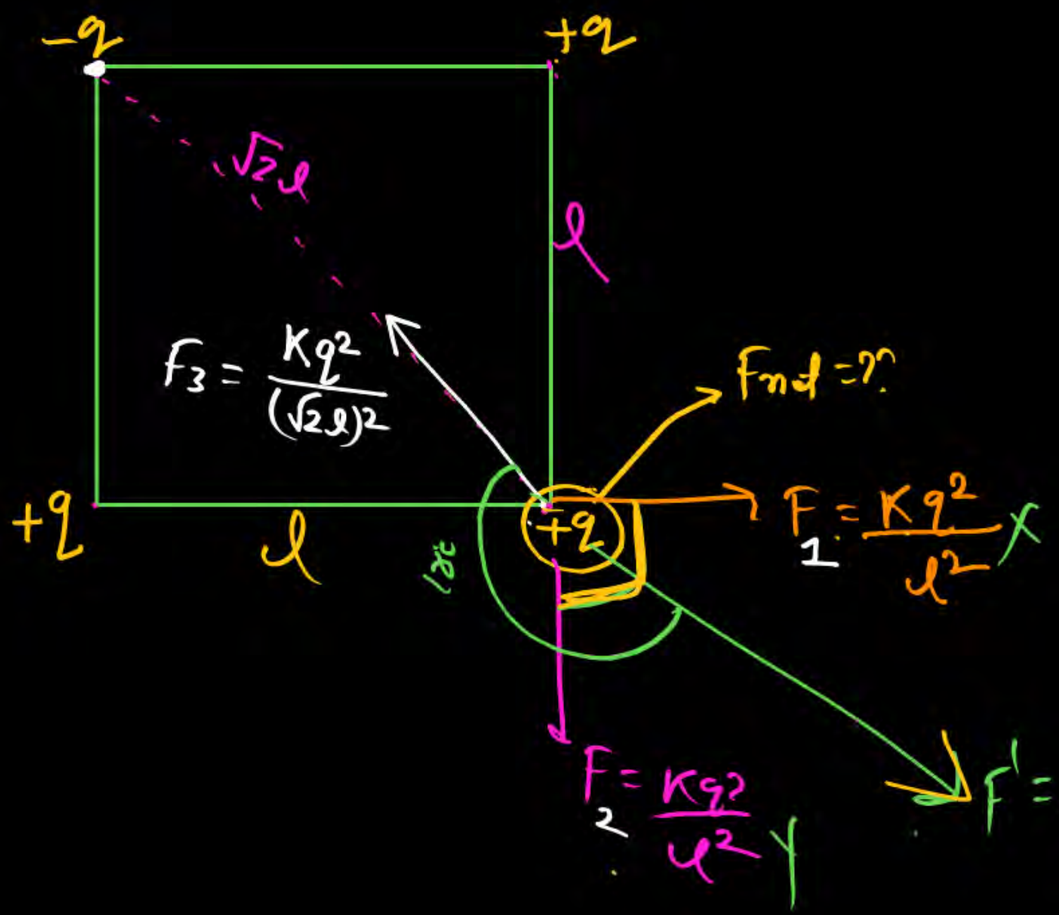
(Q) find force on any-one charge due to other 2-charge



* $F_{net} = \frac{Kq^2}{q^2}$

Q2

(a) Find net force on given charge q



$$F_{net} = \frac{\sqrt{2}Kq^2}{l^2} - \frac{Kq^2}{(\sqrt{2}l)^2}$$

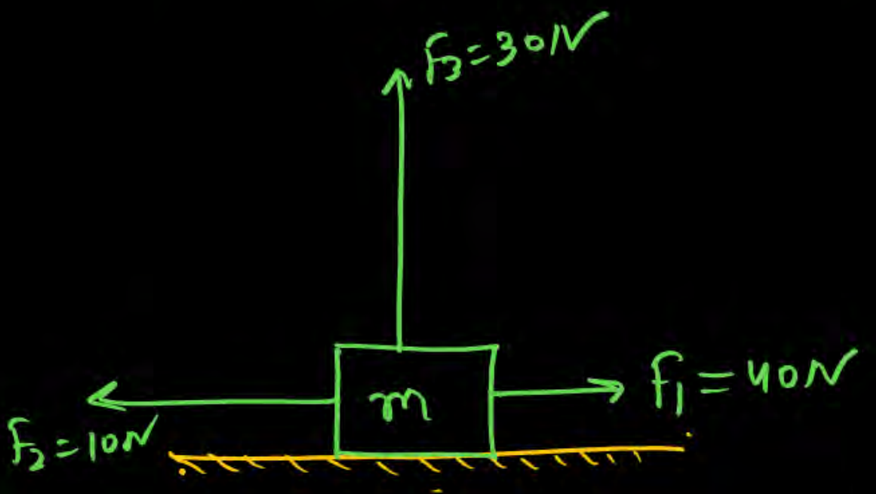
$$= \frac{Kq^2}{l^2}\left[\sqrt{2} - \frac{1}{2}\right]$$

$$= \frac{Kq^2}{l^2}\left[\frac{2\sqrt{2}-1}{2}\right]$$

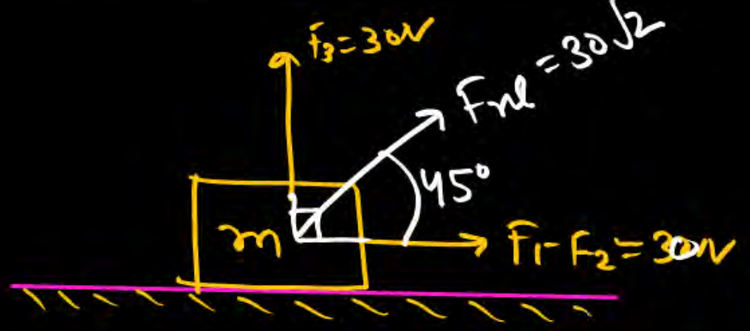
R

find F_{net} and dirⁿ of F_{net} from X-axis :- (Ignore gravity)

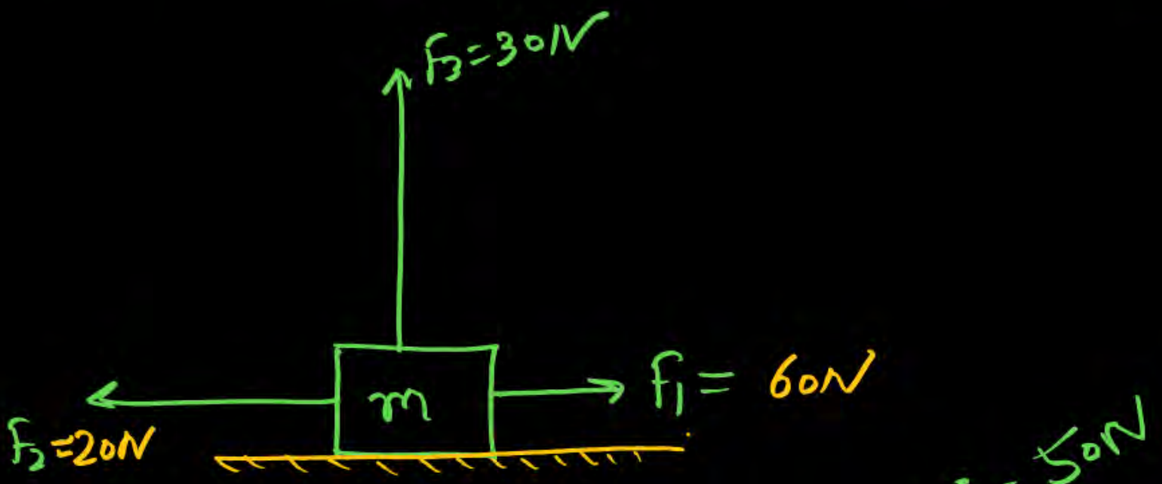
$F_{net} = F_1 + F_2 + F_3$



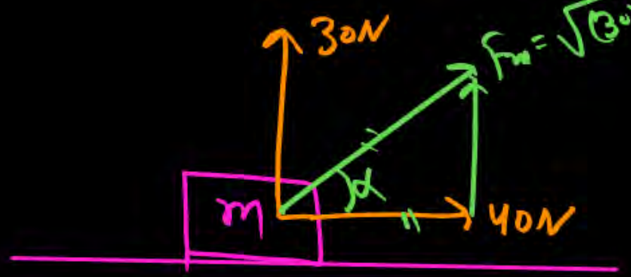
Soln



⑧ find Fnet and dirn of Fnet from X-axis :- (Ignore gravity & Normal)



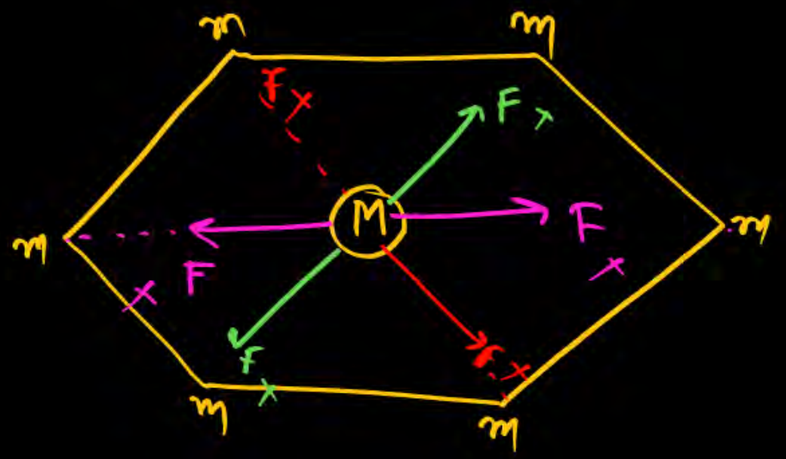
Soln



$\tan\alpha = \frac{30}{40} = \frac{3}{4}$

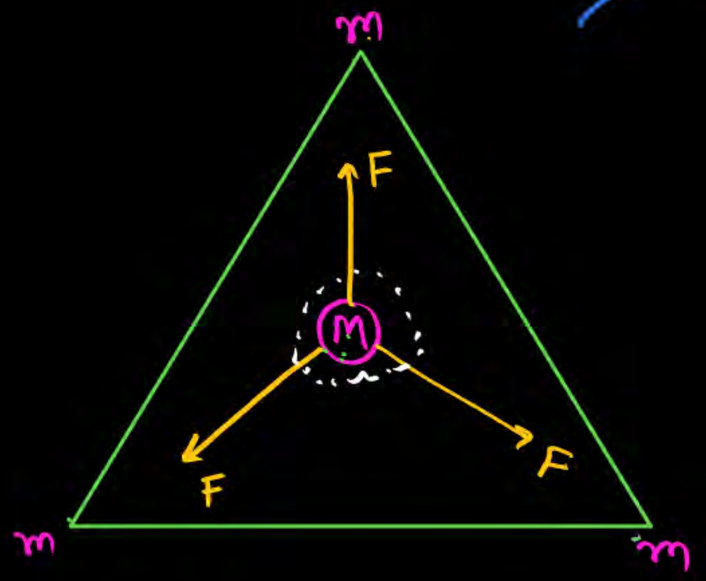
$\alpha = 37°$

Find F_{net} (gravitati.) on M which is at centre.

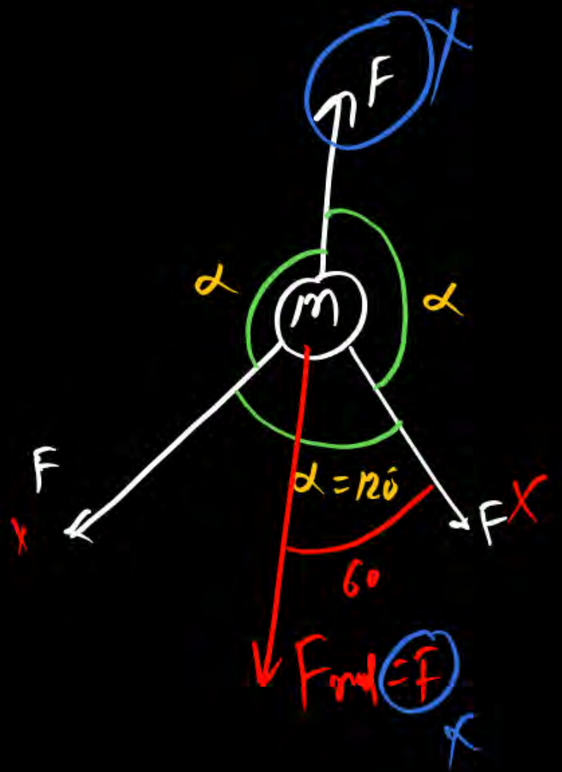


F_{net} on $m = 0$

Q Fnet on (M) mass which is at centre.



Sol^n



$3\alpha = 360$

$\alpha = 120°$

$$F = \frac{G m_1 m_2}{(d)^2}$$

$F_{net} = 0$

Question

True/false

Copy

(A) Resultant of two vector must be in the plane of two vector. (True)

(B) Minimum number of unequal vectors can give zero resultant is 2. (False)

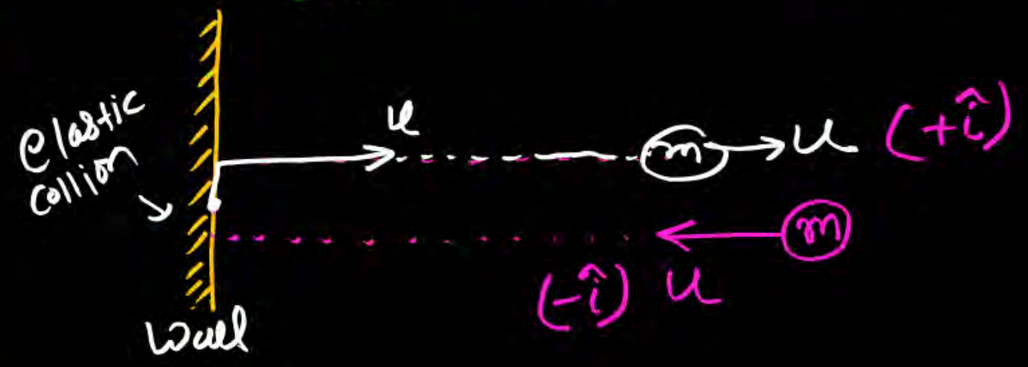
(C) Minimum number of vector can give zero resultant is 2. → True → when both vecto equal magn

(D) Minimum number of unequal vector can give zero resultant is 3. True

(E) Minimum numb of No-Coplanr Vectr can give zero resultant is 4 → True

Find change in velocity in given case

① Ball rebound with same speed



Change in speed = 0

$\vec{u}_i = -u\,\hat{i}$

$\vec{u}_f = u\,\hat{i}$

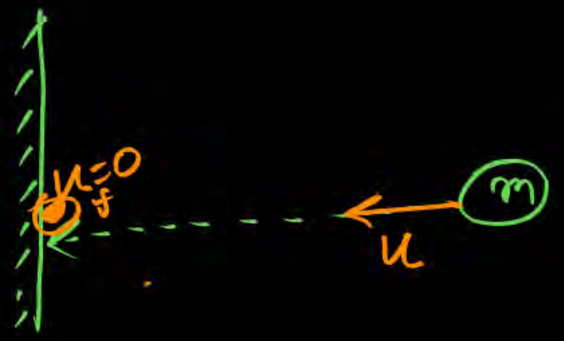
Change in velocity $= \vec{u}_f - \vec{u}_i$

$= u\hat{i} - (-u\hat{i})$

$\vec{\Delta v} = 2u\hat{i}$ ✓✓

Magnitude of change in velocity $|\vec{\Delta v}| = 2u$

② Collide and stop

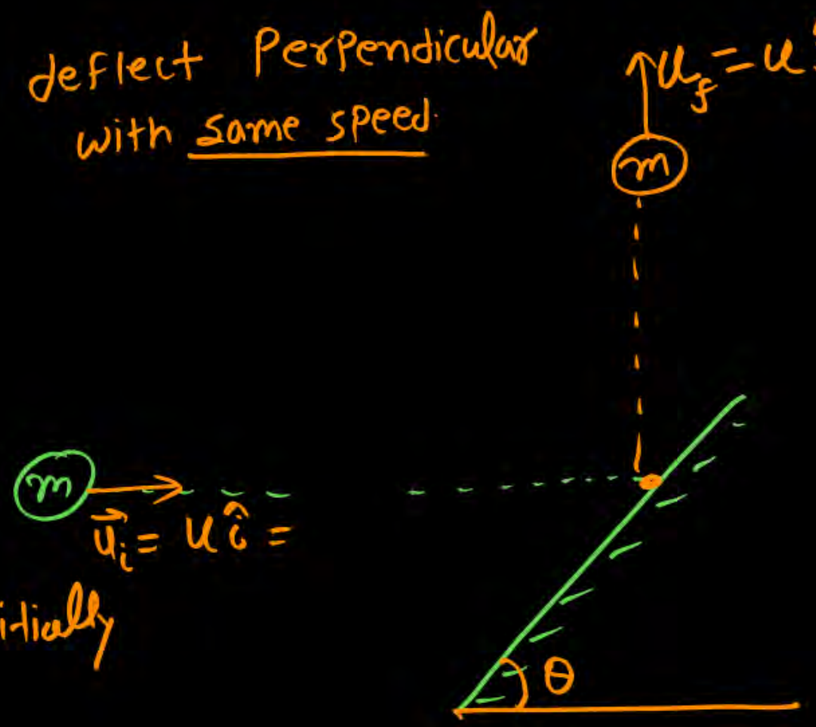


$\vec{\Delta v} = \vec{v}_f - \vec{v}_i$

$= 0 - (-u\hat{i})$

$= +u\hat{i}$ ✓

③ deflect Perpendicular with same speed



initially

Change in velocity $= \vec{u}_f - \vec{u}_i$

$= u\hat{j} - u\hat{i}$

$\vec{\Delta v} = u\hat{j} - u\hat{i}$

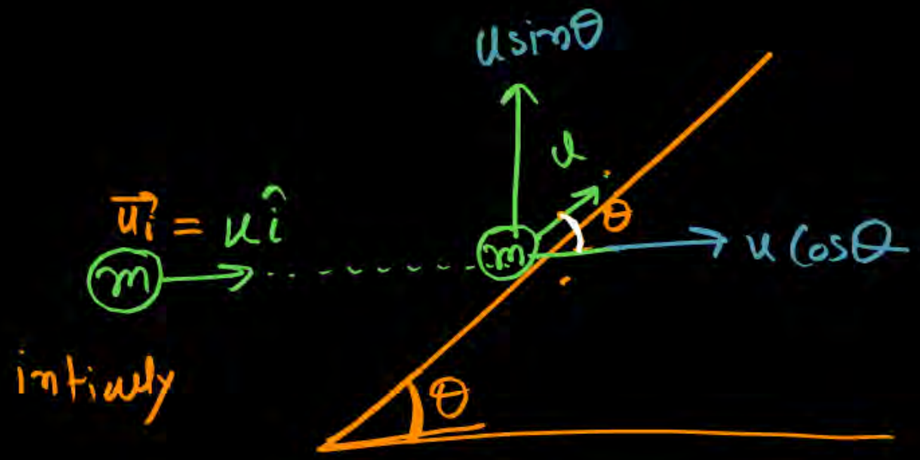
$|\vec{\Delta v}| = \sqrt{u^2 + u^2} = \sqrt{2u^2} = \sqrt{2}u$

Change in momentum

$\Delta P = \vec{P}_f - \vec{P}_i$

$= mu\hat{j} - mu\hat{i}$

④ After collision object moves with same speed along Inclined:



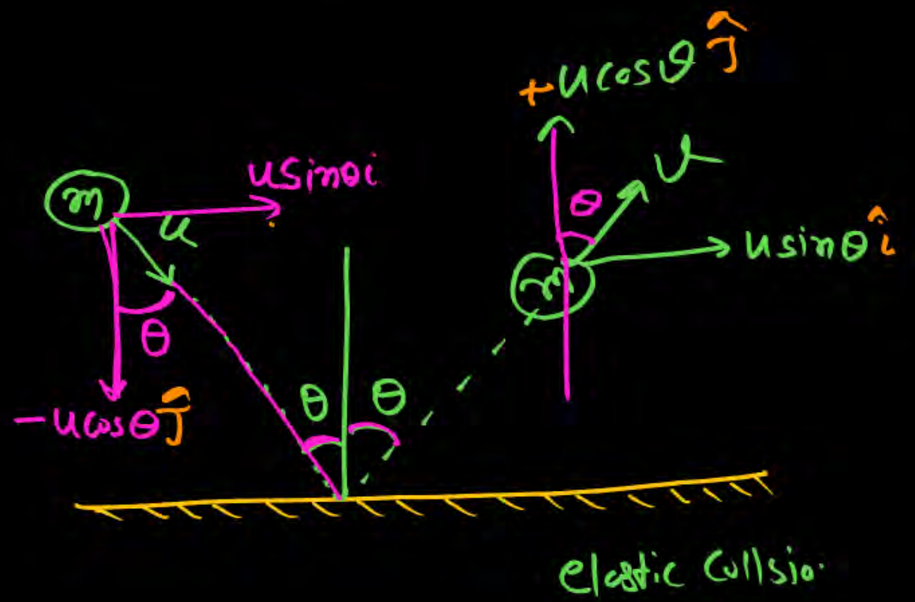
$\vec{\Delta v} = ??$

$\vec{u_i} = u\hat{i}$

$\vec{u_f} = u\cos\theta\,\hat{i} + u\sin\theta\,\hat{j}$

$\vec{\Delta v} = (u\cos\theta\,\hat{i} + u\sin\theta\,\hat{j}) - u\hat{i} = u[\cos\theta - 1]\,\hat{i} + u\sin\theta\,\hat{j}$

(6) change in velocity :-



$$\vec{u}_i = u\sin\theta\,\hat{i} - u\cos\theta\,\hat{j} \qquad \vec{u}_f = u\sin\theta\,\hat{i} + u\cos\theta\,\hat{j}$$

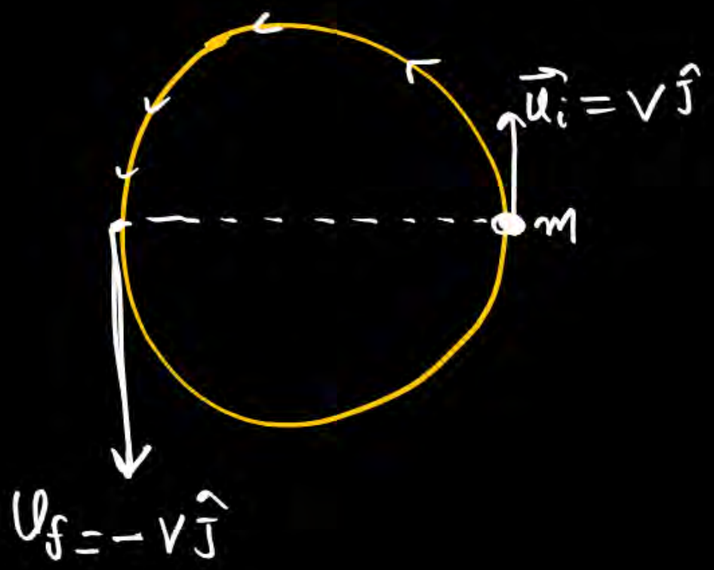
$$\Delta\vec{v} = \vec{v}_f - \vec{v}_i = (u\sin\theta\,\hat{i} + u\cos\theta\,\hat{j}) - [u\sin\theta\,\hat{i} - u\cos\theta\,\hat{j}]$$

$$= u\cos\theta\,\hat{j} + u\cos\theta\,\hat{j} = 2u\cos\theta\,\hat{j}$$

Question

(7)

Object is moving with speed V on circular path then find change in speed when it is at diametric opposite point. also find change in velocity.



change in speed $= |\vec{V}_f| - |\vec{V}_i| = 0$

" " velocity $= \vec{V}_f - \vec{V}_i$

$= -V\hat{j} - (V\hat{j})$

$\vec{\Delta V} = -2V\hat{j}$

⊙ MR* Box

Scalar can be change by changing magnitude only

but Vector can be change by changing magnitude only or direction only or by both.

Question

Initial velocity of object is 10 m/s east after some time its velocity becomes 10 m/s North then find change in velocity is?

Question

9

Majnu Majedar is moving with 6 m/s in east and Ramlal is moving with 6 m/s at 30° east of North, then find relative velocity of Ramlal w.r.t. MM.

Question

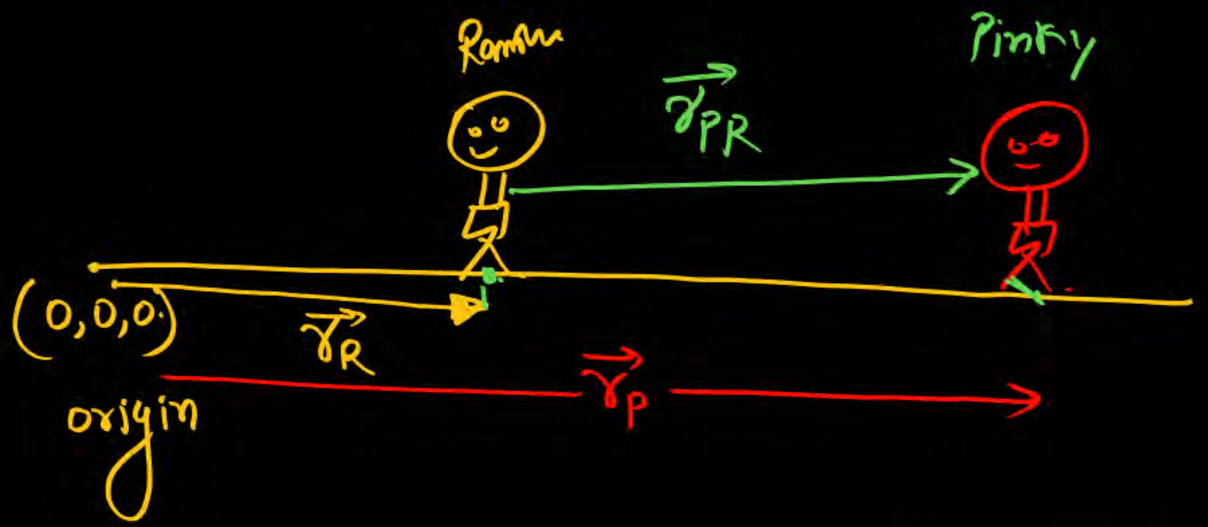
Initial velocity of Ramlal is 5 m/s in north after some time it is moving 5 m/s in east then find

(i) Change in velocity

(ii) Magnitude of change in velocity

(iii) Change in magnitude of velocity

Position vector



Position of Pinky w.r.t Ramlal. $\vec{r}_{PR} = \vec{r}_P - \vec{r}_R$

(8) find Position of object which is at (4, 3, 9) from point (1, 7, 4).

Soln



$$\vec{r}_{Positio} = (4i + 3j + 9k) - (1i + 7j + 4k)$$

$$\vec{r} = 3\hat{i} - 4\hat{j} + 5\hat{k}$$

$$|\vec{r}| = \sqrt{3^2 + (-4)^2 + (5)^2} = \sqrt{9 + 16 + 25} = \sqrt{50} = 5\sqrt{2}\,m$$

(2)



$\vec{\gamma}_{BA} = ??$

$\vec{\gamma}_{BA} = \vec{\gamma}_B - \vec{\gamma}_A = (3i + 0j + 4k) - (2i + 3j + 0k)$

$= (\hat{i} - 3\hat{j} + 4\hat{k})$ Ans

(3)

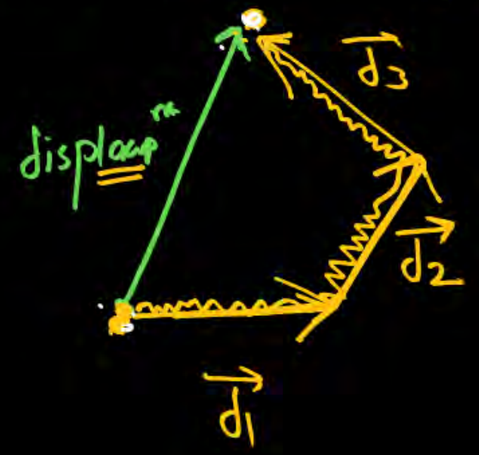


$\vec{\gamma}_{AB} = ??$

Displacement :-

$$* \left\{ \vec{d} = \vec{r}_{\text{final Position}} - \vec{r}_{\text{itial Position}} \right\}$$

(Q) If Object moves $\vec{d_1}$ then $\vec{d_2}$ then $\vec{d_3}$ then find displacement & distance



displacement $= \vec{d_1} + \vec{d_2} + \vec{d_3}$ (actual moved displ^m in vector form)

distance $= |\vec{d_1}| + |\vec{d_2}| + |\vec{d_3}|$

Question

Find position vector of A(1, 2, 1) w.r.t. point B(−1, 2, 0)?

Question JEE-main

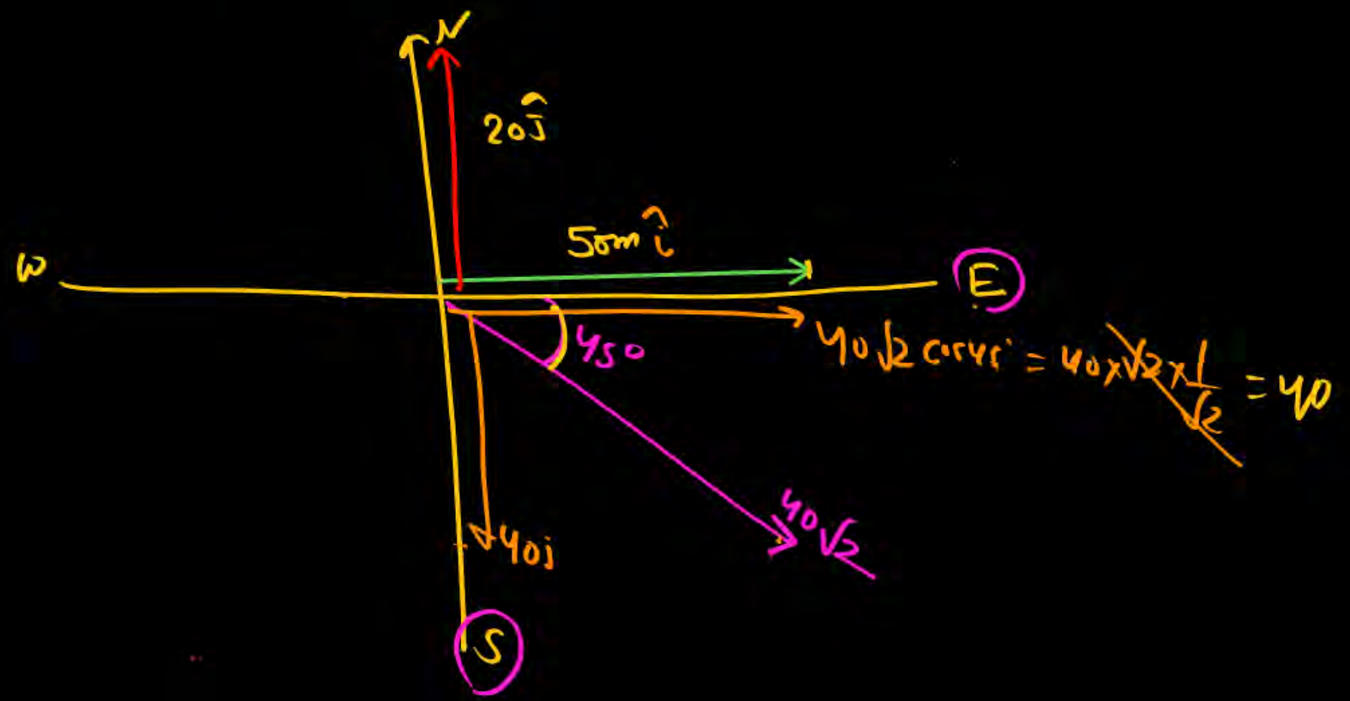
If a student moves as given below:

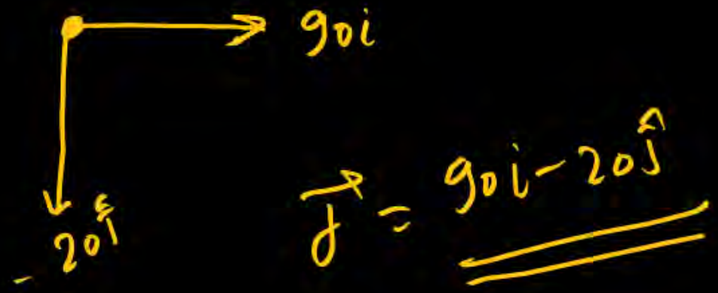
(i) 1st 50m in East

(ii) 20m in North

(iii) $40\sqrt{2}$m in S-E direction. Find net Displacement?

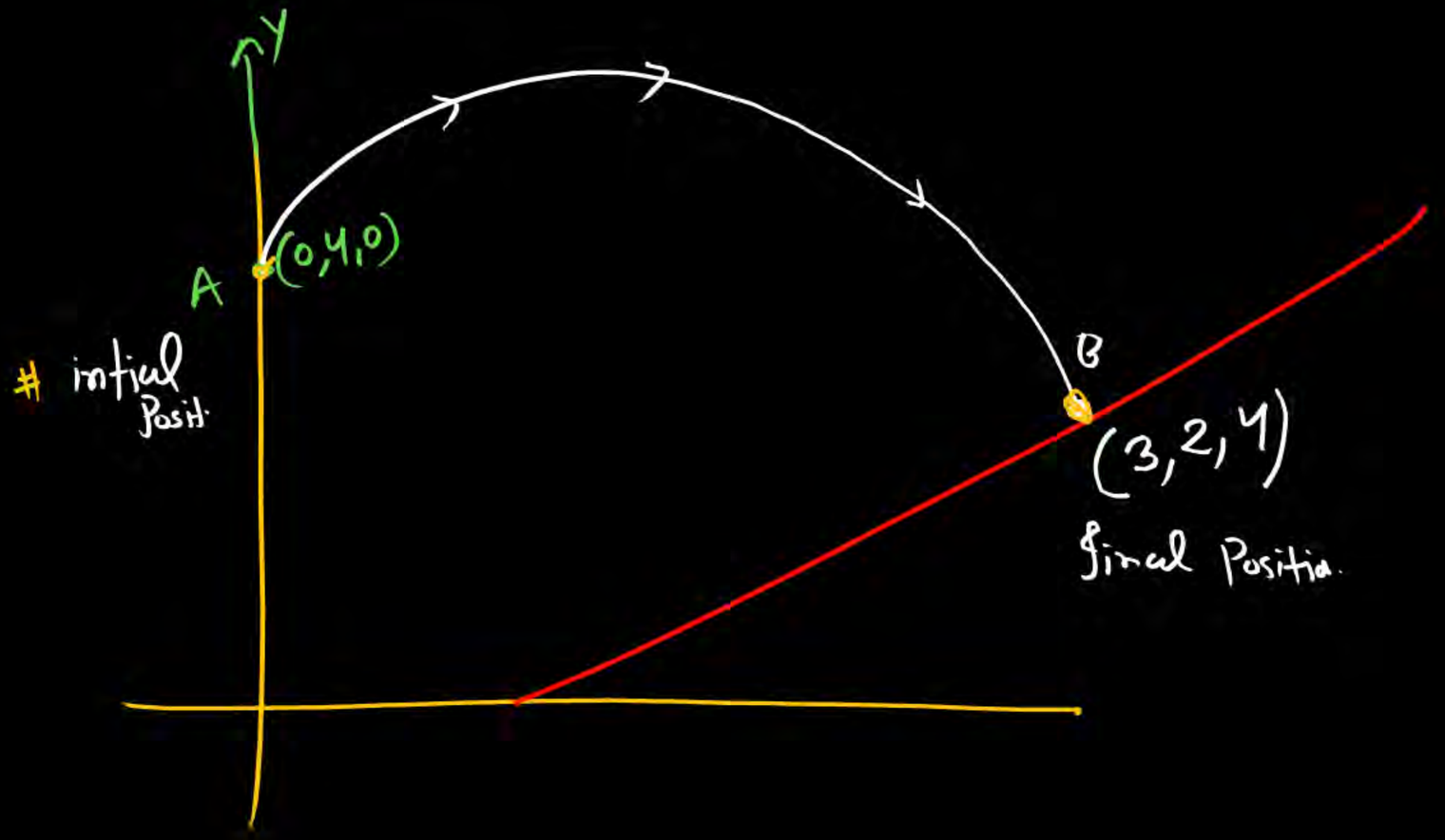
$\vec{d}_{net} = \vec{d}_1 + \vec{d}_2 + \vec{d}_3$





$\vec{d} = 90\hat{i} - 20\hat{j}$

find displ^{m} = ?



$$\text{displ}^{m} = \vec{r}_f - \vec{r}_i$$

$$= (3\hat{i} + 2\hat{j} + 4\hat{k}) - (4\hat{j})$$

$$\therefore 3\hat{i} - 2\hat{j} + 4\hat{k}$$

(a) Ramlal moves 30m in east then 50m in north then 10m in south then find displ^n.



displ^n = $30\hat{i} + 40\hat{j}$

|displ^n| = 50m

(#) distance = $|\vec{d}_1| + |\vec{d}_2| + |\vec{d}_3|$

= 30 + 50 + 10 = 90m

find dot Product

$\vec{A} = 2i + 3\hat{j} - 4\hat{k}$

$\vec{B} = 3i - 2j + 5\hat{k}$

$\vec{A} \cdot \vec{B} =$


@MRPHYSICSS

Thank You